

Numerical Methods: Eigenvalue Assembly and the Shooting Solver

Two routes to the compressible boundary-layer stability eigenvalue: the global spectral (Chebyshev collocation) assembly and the local shooting solver of `pyMack`

`pyMack` technical documentation

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1 Scope: the problem to be solved and the two routes

The companion document *Derivation of the Disturbance Equations* closes with three objects: the four linearised disturbance ODEs for $\hat{q}(y) = (\hat{u}, \hat{v}, \hat{T}, \hat{p})$, their operator form $\mathbf{L}(\alpha; y, \mathbf{D}) \hat{q} = c \mathbf{M}(\alpha; y) \hat{q}$, and the statement that, upon replacing the continuous derivative by the discretised derivative operator ($\mathbf{D} \rightarrow \mathbf{D}_1$, $\mathbf{D}^2 \rightarrow \mathbf{D}_2$), this pencil becomes the finite generalised eigenvalue problem $A(\alpha) \phi = c B(\alpha) \phi$. The present document supplies every constructive detail behind that statement: the grid, the differentiation matrices, the block-by-block assembly, the boundary-condition rows, the eigenvalue algorithms, and the identification of the physical mode. It also documents a second, independent numerical route to the same eigenvalue.

What is sought. For fixed flow parameters (M_e, Re) the disturbance equations admit non-trivial solutions only at isolated eigenvalues, and these eigenvalues are the entire objective of the computation:

- *Temporal problem:* given a real wavenumber α , find the complex phase speed c and mode shape; the temporal growth rate is $\omega_i = \alpha c_i$.
- *Spatial problem:* given a real frequency ω , find the complex wavenumber α and mode shape; the spatial growth rate is $\sigma = -\alpha_i$.

Every quantity the framework ultimately delivers is assembled from repeated eigenvalue solves: the *neutral curve* is the zero-growth locus $c_i(R, \alpha) = 0$ traced over a parameter sweep; the *N-factor* is the streamwise integral of the spatial growth rate at fixed physical frequency; and the transition estimate is the empirical correlation $N \approx 9\text{--}11$ [10, 11]. The numerical task is therefore to evaluate eigenpairs reliably, including the rejection of spurious eigenvalues, and cheaply enough to sweep them by the thousands.

Two routes. Numerical methods for this problem divide into two classical families [2]: *boundary-value methods*, which discretise the ODEs globally and obtain all eigenvalues of the resulting matrix pencil at once without prior knowledge of the spectrum, and *initial-value (shooting) methods*, which integrate the ODEs across the layer and locate one eigenvalue by enforcing the boundary conditions, requiring a good initial guess but very little memory. `pyMack` implements one of each: **Part A** is the spectral boundary-value route (Chebyshev collocation, block assembly, and dense eigensolves), used for spectrum surveys, neutral curves, and *N*-factors; **Part B** is the shooting route (`pymack/temporal_shooting.py`), the classical procedure of Mack [1] in the formulation of Özgen & Kırcah [3], used to refine and independently verify a mode found by Part A. Both return the same physical eigenvalue; their agreement is a strong check that a computed mode is physics rather than discretisation artifact. Every equation, coefficient, and default below is transcribed from the code, cited by file.

Notation. Overbars denote mean (base-flow) quantities; hats denote complex disturbance amplitudes. The continuous amplitude is $\hat{q}(y) = (\hat{u}, \hat{v}, \hat{T}, \hat{p})$; its discrete (stacked, grid-sampled) form is ϕ . A prime on a mean quantity denotes $\mathbf{D} \equiv \mathrm{d}/\mathrm{d}y$. The collocation order is N ; the grid has $n = N+1$ points, so the assembled matrices are $4n \times 4n$, the size written $4(n+1) \times 4(n+1)$ in the companion document's indexing.

Base flow & transport		Disturbance / eigenvalue	
$\bar{U}, \bar{T}, \bar{\rho}=1/\bar{T}$	mean velocity, temp., density	$\hat{q}; \phi$	amplitude (continuous; discrete)
$\bar{\mu}, \bar{\kappa}$	viscosity, conductivity	$\hat{u}, \hat{v}, \hat{T}, \hat{p}$	velocity/temp./pressure amplitudes
M_e, Re, Pr, γ	Mach, Reynolds, Prandtl, c_p/c_v	α, ω	wavenumber, frequency
$L^*, R=\sqrt{Re_x}$	Mack length, Mack Reynolds number	$c=\omega/\alpha$	complex phase speed; c_r, c_i its parts
$\nu_R=1/(Re\bar{\rho})$	viscous prefactor	$\omega_i=\alpha c_i; \sigma=-\alpha_i$	temporal; spatial growth rate
$C=\bar{\mu}/\bar{T}$	Chapman–Rubesin group	$N; n=N+1$	collocation order; grid size

Operator shorthands: D_1, D_2 differentiation matrices; $\mathcal{C}_\kappa, \mathcal{C}_d, d_R$ conduction/dissipation prefactors; $\mathcal{L}_c$ conduction operator; A, B (temporal) and C_0, C_1, C_2 (spatial) operator blocks; $Y, M(c), \sigma_{\min}(c)$ (Part B) shooting state, wall matrix, and its smallest singular value.

Part A: The spectral (Chebyshev collocation) method

The disturbance equations are four coupled ODEs in y with known base-flow coefficients but no closed-form solution. Part A converts them into a finite matrix pencil, solves the pencil for its complete spectrum, identifies the physical mode, and sweeps the solve into neutral curves and N -factors; this is the boundary-value route of Section 1.

2 Chebyshev collocation

The first task is to replace the wall-normal derivative D by a finite matrix, so that the disturbance ODEs become linear algebra and each continuous amplitude $\hat{q}(y)$ becomes the vector of its values at chosen grid points.

Nodes and spectral accuracy. For smooth functions a polynomial interpolant through Chebyshev points converges exponentially with the polynomial degree [4, 5], so few points yield many digits; this property makes spectral collocation the natural discretisation for boundary-layer eigenvalue problems. The Chebyshev–Gauss–Lobatto nodes on $[-1, 1]$ are

$$\xi_j = \cos\left(\frac{\pi j}{N}\right), \quad j = 0, 1, \dots, N, \quad (1)$$

clustering near the ends $\xi = \pm 1$. The node ordering matters throughout what follows: $\xi_0 = +1$ and $\xi_N = -1$, so the index runs from +1 down to -1.

The differentiation matrix. Exactly one polynomial of degree $\leq N$ passes through the values $\{f(\xi_j)\}$; differentiating it and evaluating at the nodes is a linear operation, $f'(\xi_j) \approx \sum_k D_{jk} f(\xi_k)$, with entries [6, 5] (`spectral.py:chebyshev_D`)

$$D_{jk} = \frac{c_j}{c_k} \frac{(-1)^{j+k}}{\xi_j - \xi_k} \quad (j \neq k), \quad D_{jj} = -\sum_{k \neq j} D_{jk}, \quad c_0 = c_N = 2, \quad c_{\text{else}} = 1. \quad (2)$$

The diagonal is set so that each row sums to zero (a constant function has zero derivative), and $c_0 = c_N = 2$ accounts for the doubled weight of the two endpoints. The matrix is dense because every node’s derivative uses every node’s value; this is why the assembled stability matrices are dense. The second derivative on $[-1, 1]$ is the matrix square, $D_\xi^2 = D_\xi D_\xi$, formed *before* the domain mapping below.

Mapping to the physical layer. The flow occupies $y \in [0, y_{\max}]$ with resolution needed at the *wall*. The algebraic map of Malik [2] (`spectral.py:map_domain`) is

$$y(\xi) = L \frac{1 + \xi}{1 - \xi + 2L/y_{\max}}, \quad L = \frac{y_{\max}}{6} \quad (\text{stretching parameter}), \quad (3)$$

which sends $\xi = +1$ (node $j=0$) to $y = y_{\max}$ (freestream) and $\xi = -1$ (node $j=N$) to $y = 0$ (wall), placing roughly half the nodes below $y \approx L$ (Figure 1). By the chain rule, with $\xi_y = 1/y_\xi$ and $\xi_{yy} = -y_{\xi\xi}/y_\xi^3$, the physical differentiation matrices are

$$D_1 = \text{diag}(\xi_y) D_\xi, \quad D_2 = \text{diag}(\xi_y^2) D_\xi^2 + \text{diag}(\xi_{yy}) D_\xi. \quad (4)$$

Note that the chain rule acts on the computational second-derivative matrix D_ξ^2 ; the physical D_2 is *not* $D_1 D_1$.

Domain height and resolution. The truncation height must clear the boundary layer by a wide margin and is therefore set as a multiple of the boundary-layer scale, $y_{\max} = m (\delta^*/L^*)$, with $m \approx 8\text{--}12$ for the wall-trapped second mode and larger ($m \approx 35\text{--}45$) for the more diffuse first mode; the bare values $y_{\max} = 6/12$ are only a fallback. The default resolution is $N = 128$ ($n = 129$). Because the method is spectral, the eigenvalue converges exponentially in N [4, 5]; at the default resolution the worked-example eigenvalue of Section 7 is already converged to many digits.

Chebyshev–Gauss–Lobatto grid and the wall-clustering map

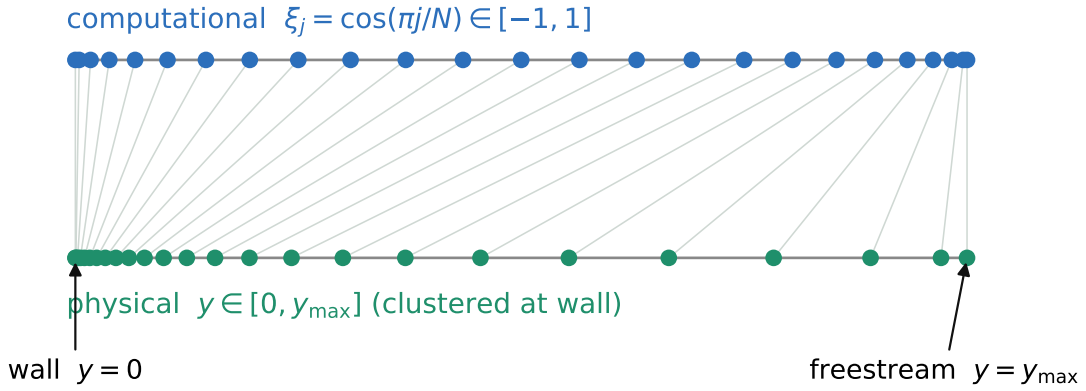


Figure 1: The Chebyshev–Gauss–Lobatto nodes $\xi_j = \cos(\pi j/N)$ on $[-1, 1]$ (top) map by Eq. (3) to the physical grid $y \in [0, y_{\max}]$ (bottom), clustering at the wall ($\xi = -1$) where the eigenfunctions vary fastest.

3 Assembling the operator and its boundary conditions

The companion document closes with the operator pencil

$$\mathbf{L}(\alpha; y, \mathbf{D}) \hat{q} = c \mathbf{M}(\alpha; y) \hat{q}, \quad \mathbf{L} = \mathbf{A}_2(y) \mathbf{D}^2 + \mathbf{A}_1(\alpha; y) \mathbf{D} + \mathbf{A}_0(\alpha; y), \quad (5)$$

with rows ordered (C, X, Y, E) and columns $(\hat{u}, \hat{v}, \hat{T}, \hat{p})$. Discretising (5) is the purely mechanical application of three substitutions on the collocation grid $\{y_j\}_{j=0}^N$ of Section 2,

$$\mathbf{D} \mapsto \mathbf{D}_1, \quad \mathbf{D}^2 \mapsto \mathbf{D}_2, \quad f(y) \mapsto \Lambda(f) = \text{diag}(f(y_0), \dots, f(y_N)), \quad (6)$$

together with the stacking of the four sampled amplitudes into the single unknown $\phi = [\hat{u}; \hat{v}; \hat{T}; \hat{p}] \in \mathbb{C}^{4n}$, the discrete eigenvector, which un-stacked is the mode shape sampled on the grid. Applied entrywise, each scalar entry of the 4×4 operator becomes an $n \times n$ block,

$$A_{ij}(\alpha) = \Lambda([A_2]_{ij}) D_2 + \Lambda([A_1]_{ij}) D_1 + \Lambda([A_0]_{ij}), \quad B_{ij}(\alpha) = \Lambda([M]_{ij}), \quad (7)$$

and the pencil becomes the generalised problem $A\phi = cB\phi$ of Section 4. Equation (7) is the entire assembly: A is built from exactly three ingredients (the differentiation matrices, diagonal samples of mean-flow profiles, and the scalars α, M_e, Re), while B , the discrete image of the algebraic operator M , marks where the eigenvalue resides. All that remains is to record the four coefficient matrices explicitly and to impose the boundary conditions.

The four coefficient matrices. Two of the four are recorded in the companion document:

$$A_2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ -\nu_R \bar{\mu} & 0 & 0 & 0 \\ 0 & -\frac{4}{3} \nu_R \bar{\mu} & 0 & 0 \\ 0 & 0 & -C_\kappa & 0 \end{pmatrix}, \quad M = i\alpha \begin{pmatrix} 0 & 0 & -1/\bar{T} & \gamma M_e^2 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \quad (8)$$

The remaining two follow by sorting every term of the four disturbance equations by derivative order, with all terms collected on the left. With the transport shorthands $m_1 = \frac{d\mu}{dT}$, $m_2 = \frac{d^2\mu}{dT^2}$, $k_1 = \frac{1}{\kappa} \frac{d\kappa}{dT}$, $k_2 = \frac{1}{\kappa} \frac{d^2\kappa}{dT^2}$ and the prefactors $\nu_R = 1/(Re\bar{\rho})$, $C_\kappa = \gamma\bar{\mu}/(Pr Re \bar{\rho})$, $C_d = \gamma(\gamma-1)M_e^2/(Re \bar{\rho})$,

$$A_1(\alpha; y) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -\nu_R \bar{\mu}' & -\frac{i\alpha}{3} \nu_R \bar{\mu} & -\nu_R m_1 \bar{U}' & 0 \\ -\frac{i\alpha}{3} \nu_R \bar{\mu} & -\frac{4}{3} \nu_R \bar{\mu}' & 0 & \bar{\rho}^{-1} \\ -2C_d \bar{\mu} \bar{U}' & (\gamma-1)\bar{\rho}^{-1} & -2C_\kappa k_1 \bar{T}' & 0 \end{pmatrix}, \quad (9)$$

$$A_0(\alpha; y) = \begin{pmatrix} i\alpha & -\bar{T}'/\bar{T} & -i\alpha \bar{U}/\bar{T} & i\alpha \gamma M_e^2 \bar{U} \\ i\alpha \bar{U} + \frac{4}{3} \alpha^2 \nu_R \bar{\mu} & \bar{U}' - i\alpha \nu_R \bar{\mu}' & -\nu_R (m_1 \bar{U}'' + m_2 \bar{U}' \bar{T}') & i\alpha \bar{\rho}^{-1} \\ \frac{2i\alpha}{3} \nu_R \bar{\mu}' & i\alpha \bar{U} + \alpha^2 \nu_R \bar{\mu} & -i\alpha \nu_R m_1 \bar{U}' & 0 \\ i\alpha(\gamma-1)\bar{\rho}^{-1} & \bar{T}' - 2i\alpha C_d \bar{\mu} \bar{U}' & i\alpha \bar{U} + C_\kappa (\alpha^2 - k_1 \bar{T}'' - k_2 \bar{T}'^2) - C_d m_1 \bar{U}'^2 & 0 \end{pmatrix}. \quad (10)$$

Equations (8)–(10), inserted into the assembly rule (7), are the complete temporal operator. They are transcribed term for term in `pymack/temporal_solver.py`, where each block $A[\text{blk}(i, j)]$ is literally (7): `np.diag` arrays multiplying D_2 , D_1 , or I . Figure 2 shows the resulting block occupancy; Figure 3 the assembled numbers for a small solved case.

Reading the matrices. The physical content is visible in the sparsity patterns:

- *Convection and the eigenvalue.* The mean convection $i\alpha \bar{U}$ occupies the $(X, \hat{u})$, $(Y, \hat{v})$, $(E, \hat{T})$ entries of A_0 and, through the state law, the $(C, \hat{T})$, $(C, \hat{p})$ entries, precisely the sparsity pattern of M . This is no coincidence: both descend from the same factor $i\alpha(\bar{U} - c)$, so B is the “ $\bar{U} \rightarrow 1$ shadow” of the convective part of A . Row C carries two such entries (the eigenvalue there multiplies the state-law pair $\hat{T}, \hat{p}$ rather than a single convected field), the other rows one each; this accounts for the five non-zero blocks of B .
- *Diffusion.* A_2 carries the only second derivatives: viscous for $\hat{u}, \hat{v}$, conductive for $\hat{T}$; its empty $\hat{p}$ column reflects the absence of a pressure diffusion mechanism.

- *First-order couplings.* In A_1 : the lone 1 at $(C, \hat{v})$ is the velocity divergence $D\hat{v}$; $(Y, \hat{p}) = \bar{\rho}^{-1}$ is the wall-normal pressure gradient; $(E, \hat{v}) = (\gamma - 1)\bar{\rho}^{-1}$ is dilatation work; the ν_R entries are viscous cross-couplings from the stress tensor; $-2\mathcal{C}_\kappa k_1 \bar{T}'$ is variable conductivity.
- *Production.* The mean-shear terms $\bar{U}'$ at $(X, \hat{v})$ and $\bar{T}'$ at $(E, \hat{v})$ in A_0 are the production terms: wall-normal motion $\hat{v}$ lifting mean momentum and mean temperature across the layer.
- *The wavenumber.* α enters only as $i\alpha$ and α^2 , so the operator is a quadratic polynomial in α ; this is exactly the property exploited in Section 5, where regrouping the same physics by powers of α yields the spatial quadratic pencil.

Example: the continuity block-row. Row C of the assembly rule (7) reads, term for term,

$$A_{(C, \hat{u})} = i\alpha I, \quad A_{(C, \hat{v})} = D_1 - \Lambda(\bar{T}'/\bar{T}), \quad A_{(C, \hat{T})} = -i\alpha \Lambda(\bar{U}/\bar{T}), \quad A_{(C, \hat{p})} = i\alpha\gamma M_e^2 \Lambda(\bar{U}),$$

$$B_{(C, \hat{T})} = -i\alpha \Lambda(1/\bar{T}), \quad B_{(C, \hat{p})} = i\alpha\gamma M_e^2 I$$

This gives six non-zero blocks, four in A and two in B , each a diagonal matrix or a diagonal matrix times D_1 . Every other block-row is read off (8)–(10) identically.

$A\phi = cB\phi$: each $4n \times 4n$ matrix is a 4×4 grid of $n \times n$ blocks (rows = equations, columns = fields)

A — dense operator (15 blocks)					B — the c -terms only (5 blocks)				
	$\hat{u}$	$\hat{v}$	$\hat{T}$	$\hat{p}$		$\hat{u}$	$\hat{v}$	$\hat{T}$	$\hat{p}$
continuity + state	ial	$D_1 - \frac{\bar{T}'}{\bar{T}}$	$-i\alpha \frac{\bar{U}}{\bar{T}}$	$i\alpha\gamma M^2 \bar{U}$	continuity + state			$-\frac{i\alpha}{\bar{T}}$	$i\alpha\gamma M^2$
x-momentum	$i\alpha \bar{U}$ + visc.	$\bar{U}'$ + visc.	transport	$i\alpha \bar{p}^{-1}$	x-momentum	$i\alpha$			
y-momentum	visc.	$i\alpha \bar{U}$ + visc.	transport	$\bar{p}^{-1} D_1$	y-momentum		$i\alpha$		
energy	dissip.	$\bar{T}' + \dots$	$i\alpha \bar{U}$ $- c_\kappa \mathcal{L}_c$	0	energy			$i\alpha$	

Figure 2: Block structure of $A\phi = cB\phi$. Rows: the four equations (continuity, x -momentum, y -momentum, energy); columns: the four fields $\hat{u}, \hat{v}, \hat{T}, \hat{p}$; shaded = non-zero block. A (left) is block-dense; B (right) holds only five non-zero blocks, the sparsity pattern of M in Eq. (8); every term there carries the eigenvalue c .

Real assembled operator: worked example ($M = 6$, $R = 5500$, $\alpha_L = 0.174$) — complements the symbolic block figure

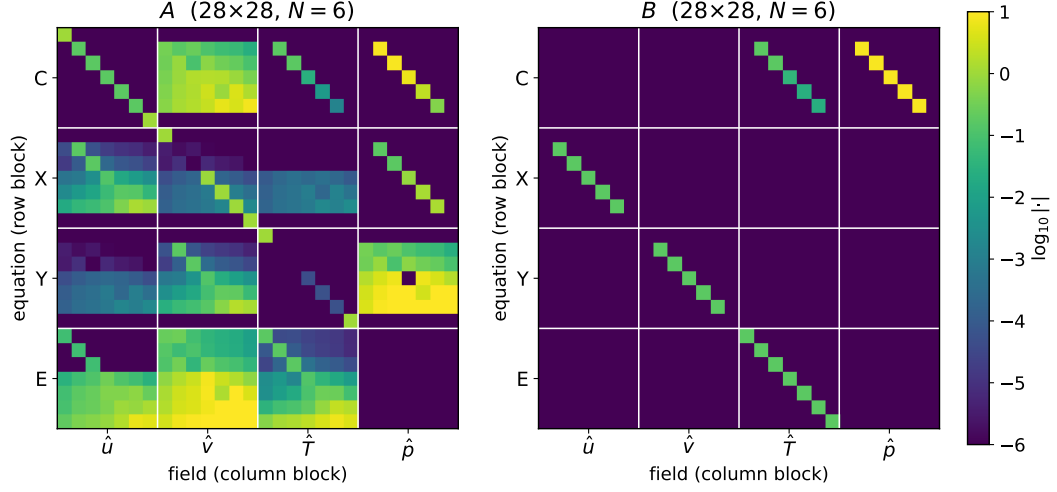


Figure 3: The same block structure assembled numerically ($N = 6$, $4n = 28$, at the parameters of Section 7), shown as $\log_{10}$ of the entry magnitudes. Gridlines every n rows/columns mark the C, X, Y, E row blocks and $\hat{u}$, $\hat{v}$, $\hat{T}$, $\hat{p}$ column blocks of Figure 2: A is densely populated throughout, while B carries only its five algebraic blocks.

Boundary conditions as row replacement. A collocation row states that the equation holds at node y_j ; at the wall and freestream those rows are overwritten by the boundary conditions. With the stacking $\phi = [\hat{u}; \hat{v}; \hat{T}; \hat{p}]$, field block $b \in \{0, 1, 2, 3\}$ occupies global indices $bn, \dots, bn + n - 1$, and, from the node ordering of Section 2, local index 0 is the *freestream* and local index $n-1$ the *wall*:

condition	field	global row	row becomes
no-slip (wall)	$\hat{u}$	$n - 1$	identity ($\hat{u} = 0$)
no-slip (wall)	$\hat{v}$	$2n - 1$	identity ($\hat{v} = 0$)
thermal (wall)	$\hat{T}$	$3n - 1$	identity ($\hat{T} = 0$, isothermal) <i>or</i> $D_1[\text{wall}, :]$ ($D\hat{T} = 0$, adiabatic)
decay (freestream)	$\hat{u}$	0	identity
decay (freestream)	$\hat{v}$	n	identity
decay (freestream)	$\hat{T}$	$2n$	identity

The same rows of B are zeroed, so each constraint is purely algebraic; these zero rows are what make B singular and the problem irreducibly generalised. The pressure $\hat{p}$ carries no explicit boundary row. The inverted node ordering is a genuine implementation hazard: because $\xi_0 = +1 \rightarrow y_{\max}$ and $\xi_N = -1 \rightarrow y = 0$, index 0 is the freestream and $n-1$ the wall, the opposite of the naive guess, and exchanging them produces plausible-looking but wrong eigenvalues (Figure 4).

Boundary conditions = row replacement
in the state vector $[\hat{u}; \hat{v}; \hat{T}; \hat{p}]$

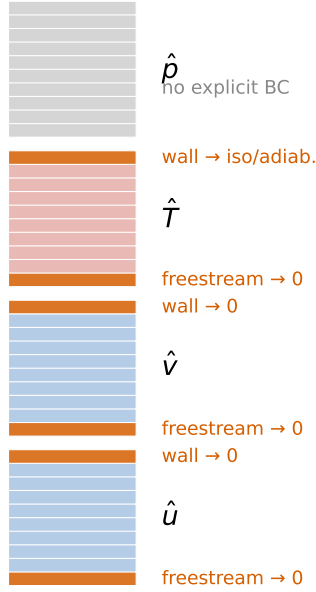


Figure 4: Boundary conditions as row replacement: the stacked state $[\hat{u}; \hat{v}; \hat{T}; \hat{p}]$ with the six overwritten rows highlighted (wall and freestream for $\hat{u}, \hat{v}, \hat{T}$); $\hat{p}$ carries no boundary row. Node index 0 is the freestream and $n-1$ the wall.

4 The temporal problem: a generalised eigenvalue problem

Given a real wavenumber α , the task is to find the complex phase speed c and mode shape ϕ ; the growth rate follows as $\omega_i = \alpha c_i$. With A, B from Section 3 this is the generalised eigenvalue problem

$$A\phi = cB\phi, \quad A, B \in \mathbb{C}^{4n \times 4n}, \quad (11)$$

solved directly by the QZ algorithm [7] (`scipy.linalg.eig`), which returns all $4n$ eigenvalues at once; no initial guess is required, the defining advantage of the boundary-value route [2]. The discrete eigenvector ϕ is the eigenfunction $\hat{q}(y)$ sampled at the grid points.

A disturbance $\propto e^{i\alpha(x-ct)}$ with real α behaves as $e^{\alpha c_i t}$ in time, so

$$c_i = \text{Im}(c) > 0 \Rightarrow \text{unstable (grows in time)}, \quad c_r = \text{Re}(c) = \text{phase speed}, \quad (12)$$

and the phase speed classifies the mode: a first (Tollmien–Schlichting) mode has $c_r \approx 0.3\text{--}0.6$, a second (Mack) mode $c_r \approx 1 - 1/M_e$ [1]. The raw spectrum is filtered to $\text{Re}(c) \in (-0.5, 1.5)$, $|\text{Im}(c)| < 0.5$ and sorted by c_i ; the identification of the physical mode within the filtered set is the subject of Section 6.

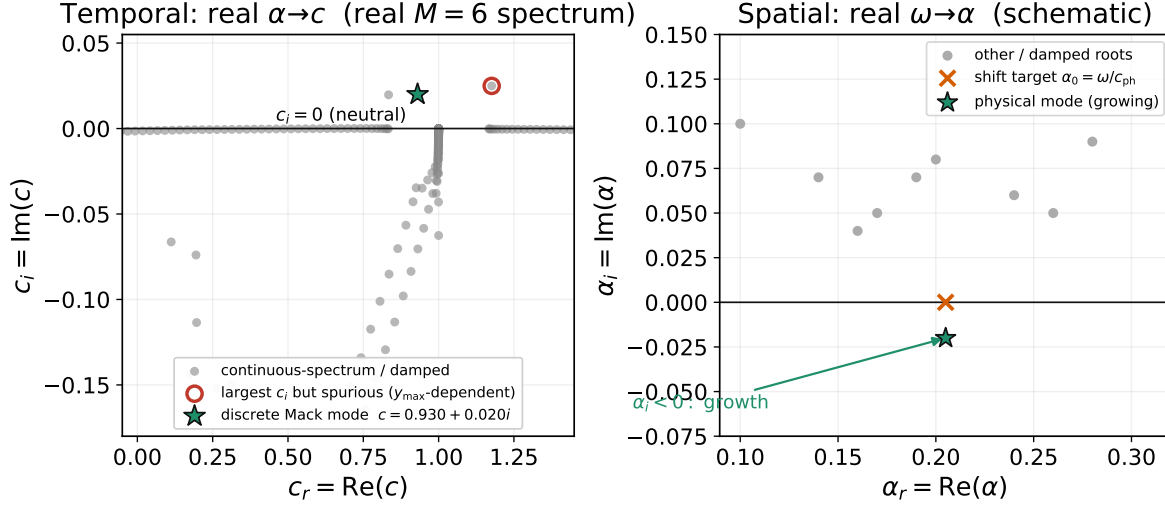


Figure 5: The two eigenvalue planes. *Left (temporal):* input real α ; the phase speeds c populate the complex plane; the physical discrete mode sits above the line $c_i = 0$ (unstable) while the continuous spectrum is damped. *Right (spatial, Section 5):* input real ω ; the wavenumbers α are sought near a shift target α_0 ; the physical mode has $\alpha_i < 0$, i.e. spatial growth.

5 The spatial problem: a quadratic eigenvalue problem

Given a real frequency ω (the physically forced case, since a fixed-frequency source generates a disturbance that grows in space), the task is to find the complex wavenumber α ; the growth rate follows as $\sigma = -\alpha_i$. Because the viscous and conduction terms carry $\partial_x^2 \mapsto -\alpha^2$, the eigenvalue now appears quadratically. Grouping the discretised equations by powers of α gives the quadratic eigenvalue problem

$$(C_0 + \alpha C_1 + \alpha^2 C_2)\phi = 0. \quad (13)$$

The blocks C_0, C_1, C_2 (complete listing). The spatial assembler (`equations.py`) uses the *enthalpy* form of the energy equation, in which $\hat{p}$ appears explicitly through Dp'/Dt and the temperature row differs from the temporal energy row, and a bulk-viscosity ratio $\lambda/\mu = 1.2$, giving the viscous coefficients of Mack [1, Eq. 8.5]

$$c_{\alpha^2} = \frac{2(2+\lambda/\mu)}{3} = 2.13, \quad c_{\times} = \frac{1+2\lambda/\mu}{3} = 1.13, \quad c_a = \frac{2(\lambda/\mu-1)}{3} = 0.13. \quad (14)$$

Because this closure is not obtained by re-grouping the temporal listing, all non-zero blocks are given, transcribed from `equations.py`; $\nu_R = 1/(Re\bar{\rho})$ is the viscous/conduction prefactor and $d_R = (\gamma - 1)M_e^2\nu_R$ the dissipation prefactor.

$$(C) \quad C_0: \hat{v} \rightarrow D_1 - \frac{\bar{T}'}{\bar{T}}, \quad \hat{T} \rightarrow \frac{i\omega}{\bar{T}}, \quad \hat{p} \rightarrow -i\omega\gamma M_e^2. \quad C_1: \hat{u} \rightarrow iI, \quad \hat{T} \rightarrow -\frac{i\bar{U}}{\bar{T}}, \quad \hat{p} \rightarrow i\gamma M_e^2 \bar{U}. \quad C_2: \text{none.}$$

$$(X) \quad C_0: \hat{u} \rightarrow -i\omega - \nu_R(\bar{\mu}D_2 + \bar{\mu}'D_1), \quad \hat{v} \rightarrow \bar{U}', \quad \hat{T} \rightarrow -\nu_R\left[\frac{d\bar{\mu}}{dT}(\bar{U}'' + \bar{U}'D_1) + \frac{d^2\bar{\mu}}{dT^2}\bar{U}'\bar{T}'\right]. \quad C_1: \hat{u} \rightarrow i\bar{U}, \quad \hat{v} \rightarrow -i\nu_R(c_{\times}\bar{\mu}D_1 + \bar{\mu}'), \quad \hat{p} \rightarrow i\bar{p}^{-1}. \quad C_2: \hat{u} \rightarrow c_{\alpha^2}\nu_R\bar{\mu}.$$

$$(Y) \quad C_0: \hat{v} \rightarrow -i\omega - c_{\alpha^2}\nu_R(\bar{\mu}D_2 + \bar{\mu}'D_1), \quad \hat{p} \rightarrow \bar{p}^{-1}D_1. \quad C_1: \hat{u} \rightarrow -i\nu_R(c_{\times}\bar{\mu}D_1 + c_a\bar{\mu}'), \quad \hat{v} \rightarrow i\bar{U}, \quad \hat{T} \rightarrow -i\nu_R\frac{d\bar{\mu}}{dT}\bar{U}'. \quad C_2: \hat{v} \rightarrow \nu_R\bar{\mu}.$$

$$(E) \quad C_0: \hat{u} \rightarrow -2d_R\bar{\mu}\bar{U}'D_1, \quad \hat{v} \rightarrow \bar{T}', \quad \hat{T} \rightarrow -i\omega - \nu_R[\bar{\kappa}D_2 + 2\bar{\kappa}'D_1 + \frac{d\bar{\kappa}}{dT}\bar{T}'' + \frac{d^2\bar{\kappa}}{dT^2}(\bar{T}')^2] - d_R\frac{d\bar{\mu}}{dT}(\bar{U}')^2, \quad \hat{p} \rightarrow i\omega(\gamma-1)M_e^2\bar{p}^{-1}. \quad C_1: \hat{v} \rightarrow -2id_R\bar{\mu}\bar{U}', \quad \hat{T} \rightarrow i\bar{U}, \quad \hat{p} \rightarrow -i(\gamma-1)M_e^2\bar{U}\bar{p}^{-1}. \quad C_2: \hat{T} \rightarrow \nu_R\bar{\kappa}.$$

All other blocks are zero. The temporal closure (Stokes $\frac{4}{3}, \frac{1}{3}$ factors, temperature energy row) and this spatial closure (enthalpy row, $\lambda/\mu = 1.2$) differ only in the bulk-viscosity treatment and the energy-equation arrangement; both approximate the same physical mode and agree on it to within discretisation error.

Companion linearisation. A quadratic pencil becomes an ordinary generalised problem of twice the size by introducing the auxiliary variable $\psi = \alpha\phi$ [8] (`solver.py`):

$$\underbrace{\begin{bmatrix} -C_1 & -C_0 \\ I & 0 \end{bmatrix}}_{\mathcal{L}} \begin{bmatrix} \psi \\ \phi \end{bmatrix} = \alpha \underbrace{\begin{bmatrix} C_2 & 0 \\ 0 & I \end{bmatrix}}_{\mathcal{R}} \begin{bmatrix} \psi \\ \phi \end{bmatrix}. \quad (15)$$

The lower block row enforces $\psi = \alpha\phi$; the upper row then reproduces (13). The eigenvalue is α and the physical shape is the lower half ϕ : since $\psi = \alpha\phi$ the two halves are parallel, and by convention the lower half is normalised as the eigenfunction (Figure 6).

Companion linearisation: a quadratic EVP becomes an ordinary EVP of double size

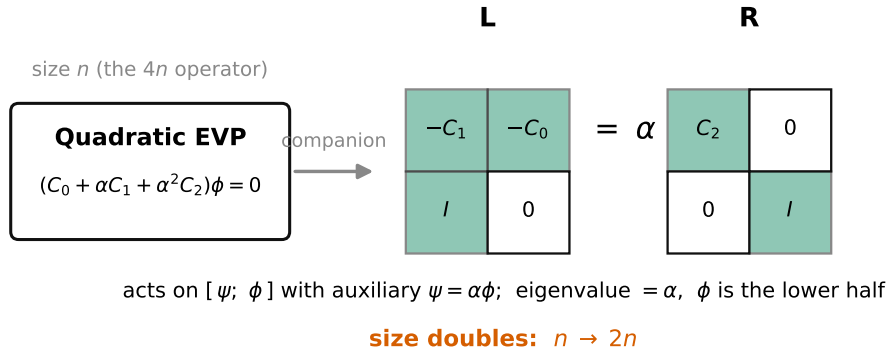


Figure 6: Companion linearisation (15): the $4n$ -sized quadratic problem becomes an ordinary generalised problem of size $8n$ in $[\psi; \phi]$ with $\psi = \alpha\phi$. Its eigenvalue is exactly α ; the eigenfunction is the lower half.

Shift–invert. The physically relevant branch is targeted with a shift $\alpha_0 = \omega/c_{\text{ph}}$ ($c_{\text{ph}} = 1 - 1/M_e$ for $M_e > 3$, else ≈ 0.4 ; these are the second- and first-mode phase-speed estimates of Section 4). Subtracting α_0 and inverting maps eigenvalues near α_0 to large μ :

$$\text{eigenvalues of } (\mathcal{L} - \alpha_0 \mathcal{R})^{-1} \mathcal{R} \text{ are } \mu = \frac{1}{\alpha - \alpha_0}, \quad \Rightarrow \quad \alpha = \alpha_0 + \frac{1}{\mu}, \quad (16)$$

so the largest- $|\mu|$ modes are the wavenumbers nearest the physical one (in `pyMack`, a dense solve followed by selecting the $n_{\text{modes}}=20$ eigenvalues nearest α_0 and sorting by $\text{Im } \alpha$). The amplitude varies as $e^{i\alpha x} = e^{i\alpha_r x} e^{-\alpha_i x}$, so

$$\sigma = -\text{Im}(\alpha) = -\alpha_i > 0 \Rightarrow \text{unstable (grows downstream)}. \quad (17)$$

Alternative route: Newton iteration from a temporal seed. When shift–invert struggles, a more robust path (`solve_spatial_from_temporal`) solves the *temporal* problem at a real trial

wavenumber $\alpha_r = \omega/c_{\text{est}}$ ($c_{\text{est}} = 0.98$ for $M_e > 3$, else 0.35), keeping only modes that agree across two resolutions (tolerance 8×10^{-3}); forms the seed $\text{Im}(\alpha) \approx -\alpha_r c_i / c_r$, a phase-speed estimate of the rigorous group-velocity conversion of Gaster [9] that suffices here because it only seeds the iteration, and then Newton-iterates the dispersion residual $F(\alpha) = \alpha c(\alpha) - \omega = 0$, $\alpha^{(k+1)} = \alpha^{(k)} - F/F'$, with F' approximated by a finite difference (step 10^{-5} , tolerance 10^{-6} , at most 20 iterations). Both routes return the same $\sigma = -\alpha_i$.

6 Identifying the physical mode

A single solve of (11) returns $4n$ eigenvalues, of which only a handful are genuine boundary-layer modes; the task is to select them. Besides the discrete modes, the operator possesses a *continuous spectrum*: on an unbounded domain the freestream solutions form a continuous band rather than isolated points, which the finite grid renders as a dense cluster of spurious discrete eigenvalues. Two tests (`discrete_mode.py`) separate the physical mode:

1. **Freestream decay.** A true boundary-layer eigenfunction decays as $y \rightarrow y_{\text{max}}$; a continuous-spectrum mode oscillates undiminished to the edge. Candidates whose edge-to-peak amplitude ratio exceeds ~ 0.06 are rejected.
2. **Domain stationarity.** A genuine eigenvalue is insensitive to the truncation height. The solve is repeated at two domain heights and only modes whose c matches to $\sim 4 \times 10^{-3}$ are kept; accepted growth rates additionally satisfy $|c_i| \lesssim 0.05$ with c_r in the physical band.

These two tests are what allow a single trusted number to be extracted from the full spectrum; the worked example below shows why they are indispensable: its largest-growth eigenvalue is in fact spurious.

7 Worked example: one temporal solve

Consider a strong Mack-mode case: $M_e = 6.0$, $R = 5500$, $\alpha_L = 0.174$, adiabatic wall, L^* scaling, $N = 180$ ($n = 181$, matrices $4n = 724$ wide), and domain height $y_{\text{max}} = 10(\delta^*/L^*) \approx 128$ (a multiple of the boundary-layer scale, per Section 2). Solving (11) returns 724 eigenvalues; after the physical filter, the six least-stable are:

c_r	c_i	$\omega_i = \alpha c_i$	y_{max} -stationary?	verdict
+1.176	+0.0249	+0.00434	no ($ \Delta c \sim 10^{-2}$)	spurious / continuous spectrum
+0.930	+0.0200	+0.00348	yes ($ \Delta c \sim 10^{-7}$)	discrete Mack mode (the answer)
+0.834	+0.0197	+0.00343	no	continuous spectrum
+0.740	+0.0000	0.00000	n/a	neutral continuous-spectrum cluster
+0.720	+0.0000	0.00000	n/a	continuous spectrum
+0.760	+0.0000	0.00000	yes	stable discrete mode

This is precisely the trap the tests of Section 6 exist to avoid: the largest-growth eigenvalue (first row, $c_r = 1.18$) is *not* the answer: its c shifts by $\sim 10^{-2}$ when the domain height changes, and $c_r > 1$ is unphysically supersonic. The physical answer is the second row, the Mack (second) mode $c = 0.930 + 0.0200i$: domain-stationary to 10^{-7} , with an eigenfunction that decays at the freestream (Figure 7). Its growth rate is $\omega_i = \alpha c_i = +0.00348$ (unstable) and its phase speed $c_r = 0.93$ is the second-mode value near $1 - 1/M_e$ [1]. This single $\omega_i > 0$ is one point inside the neutral curve; repeating the solve across an (R, α) grid and locating the zero crossing of ω_i traces the neutral curve of Section 8. All numbers reproduce from `solve_temporal_2d`.

Computed eigenfunction: $M = 6$ Mack mode, $c = 0.930 + 0.020i$

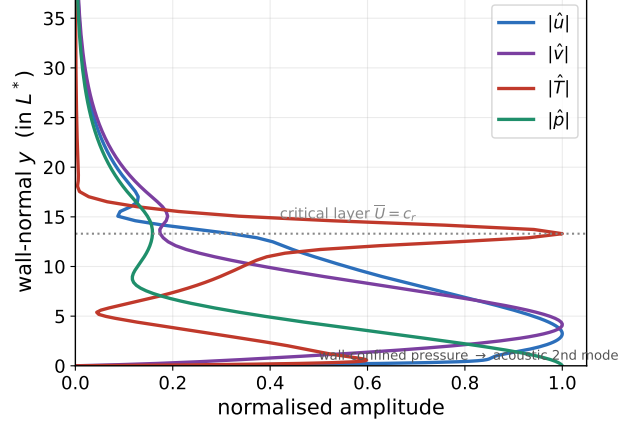


Figure 7: Computed eigenfunction of the worked-example Mack mode: the four amplitude fields $|\hat{u}|$, $|\hat{v}|$, $|\hat{T}|$, $|\hat{p}|$ against y (each normalised). $|\hat{T}|$ peaks at the critical layer ($\bar{U} = c_r$); $|\hat{p}|$ is wall-confined, the acoustic signature of the second mode. All fields decay at the freestream, confirming a discrete mode.

8 Neutral curves

The neutral curve is the boundary between growth and decay: the locus $c_i(R, \alpha) = 0$ (temporal) or $\sigma(R, \omega) = 0$ (spatial), inside which the flow is unstable. `pyMack` traces it in three steps:

1. **Sweep** an (R, α) grid at fixed M_e ; at each node solve (11), apply the mode tests of Section 6, and store c_i (`build_*_grid.py`).
2. **Extract** the zero contour of the stored field by marching squares (`extract_contours.py`); unresolved nodes are tagged stable so that the contour closes.
3. **Continuation** for difficult branches: at low Mach number the first-mode eigenvalue approaches the continuous spectrum and the grid sweep loses it. `trace_continuation.py` tracks the mode by proximity: from a clean seed it marches in R , follows the nearest eigenvalue, and secant-solves $c_i = 0$ at each step, each converged eigenvalue seeding the next parameter value.

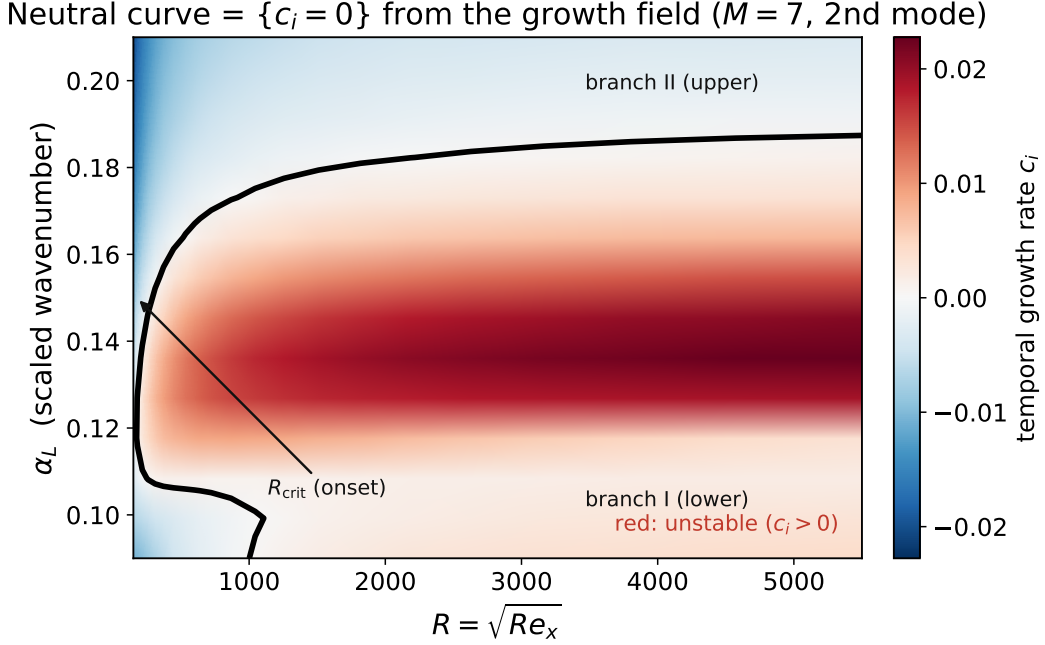


Figure 8: Steps 1–2 of the neutral-curve procedure for a solved case ($M_e = 7$, second mode): the coloured field is the growth rate $c_i(R, \alpha)$ from the sweep; the black curve is the extracted $\{c_i = 0\}$ contour. The flow is unstable inside the loop; R_{crit} marks the lower branch (branch I) where instability first appears as R increases, and the curve closes at branch II.

9 N -factor and transition

What matters for transition is the *accumulated* amplification of a fixed-frequency disturbance. For a chosen reduced frequency F , the spatial problem of Section 5 is solved at successive stations to obtain $\sigma_L = -\text{Im}(\alpha_L)$, the spatial growth rate (17) expressed in the L^* scaling, and integrated from the lower neutral point R_0 where σ_L first becomes positive:

$$N(R) = \int_{R_0}^R \max(\sigma_L, 0) dR' \quad (\text{trapezoidal}), \quad \frac{A(R)}{A(R_0)} = e^{N(R)}. \quad (18)$$

The amplitude grows by e^N , and transition is empirically expected near $N \approx 9$ –11, following the e^N method of Smith & Gamberoni and van Ingen [10, 11] (`analysis.py`). For a cone, surface arc length is mapped to an equivalent R before integration. The transition N is a correlation, not a first-principles criterion: it depends on the disturbance environment (flight ~ 9 –11; noisy conventional tunnels transition at $N \sim 4$ –6), presumes a single dominant linear mode, and absorbs the unknown initial amplitude set by receptivity.

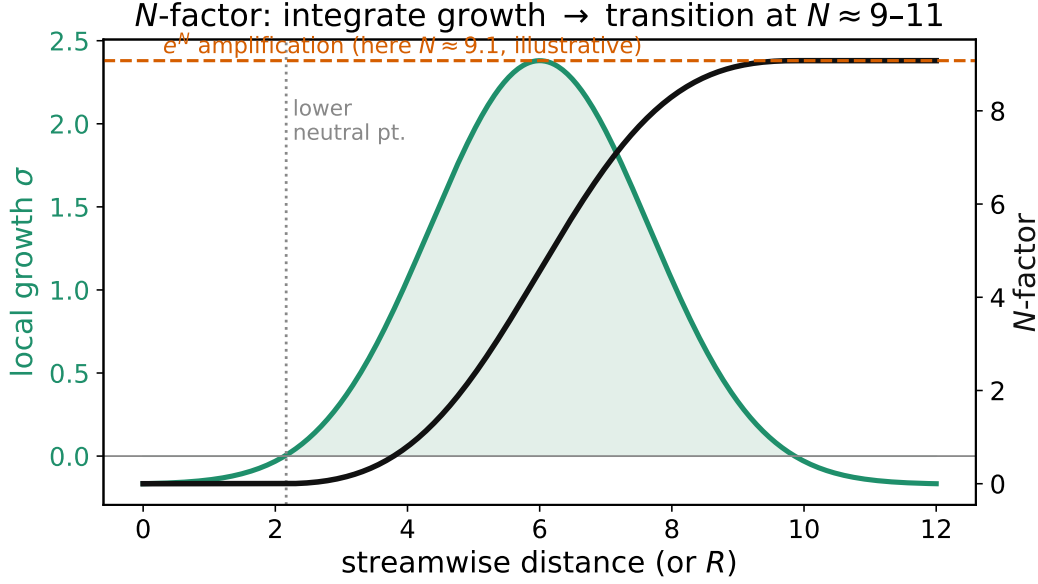


Figure 9: The N -factor integration, Eq. (18): the shaded area is $\max(\sigma_L, 0)$, the local spatial growth carried from station to station; the black curve is the running integral $N(R)$. The growth-rate curve is schematic (a rescaled synthetic $\sigma_L(R)$), drawn to reach the empirical transition band $N \approx 9\text{--}11$ so the asymptote can be read against that criterion.

10 Part A summary and replication checklist

Goal	Problem solved	Method	Growth
Temporal growth, neutral curves	generalised EVP $A\phi = cB\phi$, real α (Eq. 11)	Chebyshev collocation + QZ	αc_i
Spatial growth, N -factor	quadratic EVP $(C_0 + \alpha C_1 + \alpha^2 C_2)\phi = 0$, real ω (Eq. 13)	companion + shift-invert; or temporal + Newton seed	$-\alpha_i$
Neutral curve	$\{c_i = 0\}$ over an (R, α) sweep (Figure 8)	grid sweep $\rightarrow$ marching squares; continuation	n/a
Transition	$N = \int \max(\sigma_L, 0) dR$ (Eq. 18, Figure 9)	fixed- F spatial march + integration	n/a

To re-implement the spectral solver from scratch.

1. Solve the base-flow boundary-value problem (companion *Mean Boundary-Layer* document); tabulate $\bar{U}, \bar{T}$ and derivatives.
2. Build ξ_j (1), D_ξ (2), map to y (3), form D_1, D_2 (4).
3. Assemble the 4×4 block operator (temporal: the coefficient matrices (8)–(10) via the assembly rule (7); spatial: C_0, C_1, C_2).
4. Overwrite the wall/freestream rows with the boundary conditions (Section 3), respecting the node ordering.
5. Solve the eigenvalue problem (temporal (11) or spatial (15)); filter for the discrete mode (Section 6); read c_i or $-\alpha_i$.

6. Sweep parameters; extract $\{c_i = 0\}$ for neutral curves, or integrate $-\alpha_i$ for N -factors.

Code map (Part A). spectral operators: `pymack/spectral.py`; temporal assembly: `pymack/temporal_solver.py`; spatial assembly: `pymack/equations.py`; solvers / companion / Newton: `pymack/solver.py`; mode selection: `discrete_mode.py`. Neutral-curve drivers and continuation: `build*_grid.py`, `extract_contours.py`, `trace_continuation.py`.

Part B: The shooting solver

Part A built one large matrix and asked a dense eigensolver for all of its eigenvalues. Part B poses the identical temporal question (for a given real α , at what complex c does a non-trivial disturbance satisfy decay at the freestream and the wall conditions?) as an *initial-value* computation [2]: the disturbance equations are recast as a small first-order system, integrated from the freestream to the wall, and the eigenvalue is located by driving a 3×3 wall matrix singular. No global matrix is ever formed; the cost per evaluation is small, but a good initial guess is required. This is the classical shooting procedure of Mack [1], implemented in `pymack/temporal_shooting.py` in the formulation of Özgen & Kırcah [3]. Within `pyMack` its role is verification and refinement: given a mode located by the spectral solver, shooting confirms it through entirely independent numerics and sharpens it to tight tolerance. Figure 10 shows the complete pipeline.

Part B shooting scheme: eigen-basis $\rightarrow$ RK4+QR march $\rightarrow$
wall matrix $\rightarrow$ Nelder-Mead search

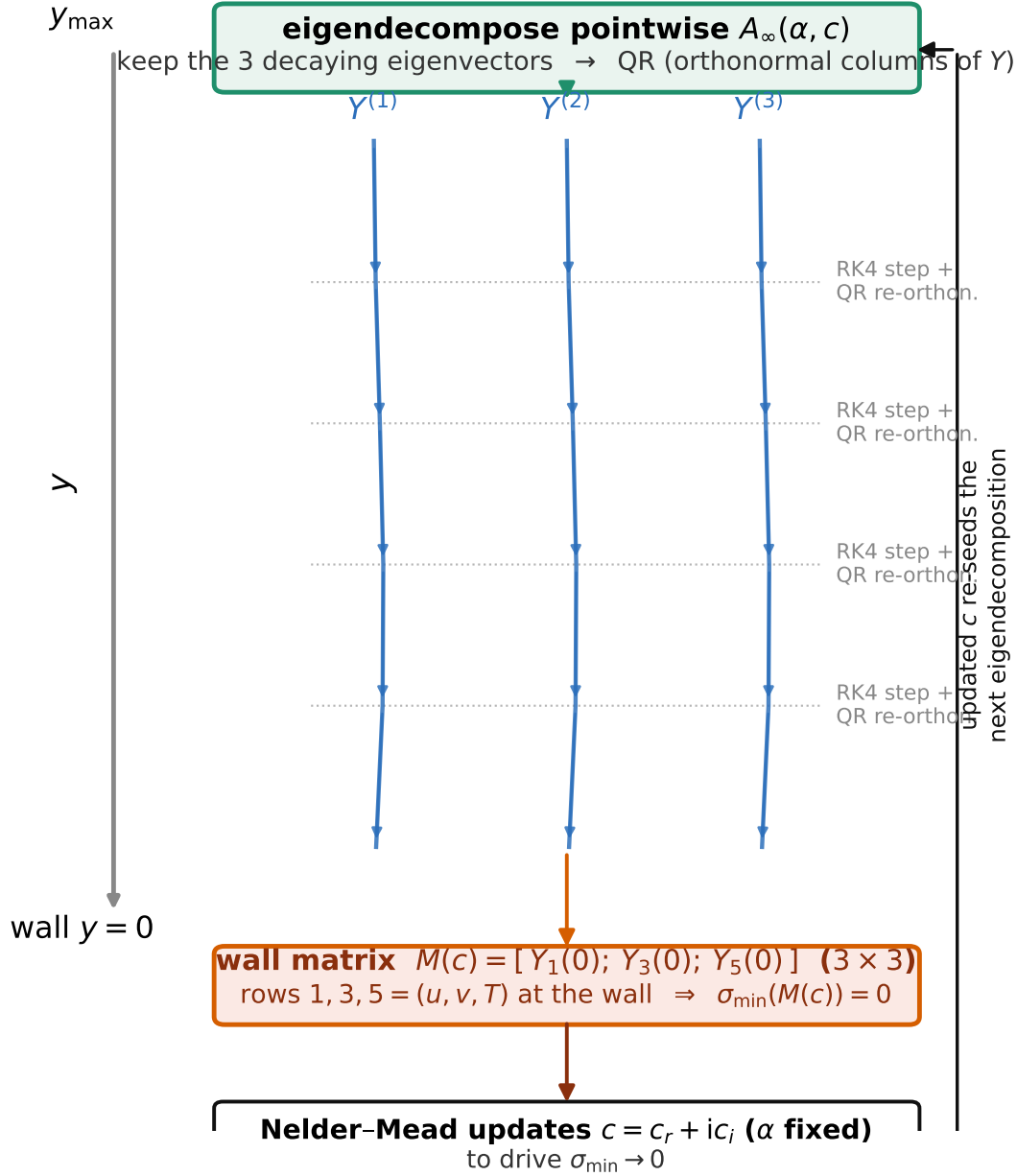


Figure 10: The shooting pipeline, top ($y_{\max}$) to bottom (wall, $y = 0$). At $y_{\max}$ the constant-coefficient matrix A_{∞} is eigen-decomposed and its three decaying eigenvectors are QR-orthonormalised into the starting 6×3 basis (Section 12); the basis is RK4-marched to the wall with re-orthonormalisation after every step (Section 13); at the wall the three arriving columns form the 3×3 matrix $M(c)$ (Section 14); and an outer Nelder-Mead loop (Section 15) updates the trial c to drive $\sigma_{\min}(c) \rightarrow 0$, feeding back into the eigen-decomposition at $y_{\max}$.

11 The first-order state and its coefficient matrix

Given a real wavenumber α and a trial phase speed c , the four disturbance equations are cast as a single first-order ODE in a six-component complex state. Following Özgen & Kırçalı [3, Eqs. 22–28], specialised to two dimensions ($\beta = 0$, no spanwise velocity), the state is

$$Y(y) = (X_1, X_2, X_3, X_4, X_5, X_6)^\top \in \mathbb{C}^6, \quad (19)$$

in which X_1, X_3, X_5 are the primary flow amplitudes, X_2, X_6 the wall-normal derivatives of X_1, X_5 , and X_4 the pressure amplitude. Concretely, $X_1 = \alpha \hat{u}$ is the wavenumber-scaled streamwise-velocity amplitude (the k -direction combination $\alpha \hat{u} + \beta \hat{v}$ of [3] evaluated at $\beta = 0$; the constant factor α is immaterial in a homogeneous eigenvalue problem), $X_3 = \hat{v}$ the wall-normal velocity, $X_5 = \hat{T}$ the temperature (with $X_2 = DX_1$, $X_6 = DX_5$), and $X_4 = \hat{p}$ the same $\rho_e U_e^2$ -scaled pressure as Part A, coupled through the continuity and y -momentum rows rather than as a plain derivative. The state thus stacks $(\alpha u, \alpha u', v, p, T, T')$, and the system closes as one first-order ODE,

$$\frac{dY}{dy} = A(y; \alpha, c, Re, Me) Y, \quad A \in \mathbb{C}^{6 \times 6}, \quad (20)$$

where A is assembled *pointwise*, one 6×6 matrix at each y , from the local mean-flow data $\bar{U}, \bar{U}', \bar{U}'', \bar{T}, \bar{T}', \bar{\mu}$ with $d\mu/dT$, $d^2\mu/dT^2$, and $\bar{\kappa}$ with its temperature derivatives (function `ozgen_first_order_matrix_2d`). Rows 1 and 5 of A carry the trivial derivative definitions $DX_1 = X_2$ and $DX_5 = X_6$ (single entries $A_{12} = A_{56} = 1$); the physics resides in rows 2, 4, 6 (the two momentum balances and the energy balance) and in row 3, continuity with the equation of state.

The pointwise structure mirrors the block operator of Part A term for term: the convective factor $\alpha(\bar{U} - c)$, the viscous prefactor $\propto Re/(\bar{\mu}\bar{T})$, the transport-derivative couplings $\frac{d\mu}{dT}\bar{U}'$, and the conduction group in $\bar{\kappa}$ all reappear, now as scalar entries of a 6×6 matrix at one height rather than as $n \times n$ blocks. The gas defaults ($\gamma = 1.4$) and the length scaling (`L.star`) match the spectral solver, so both routes see the identical mean flow.

Remark: single source of truth, and a correction to the printed reference. Internally, `ozgen_first_order_matrix_2d` delegates to the Mack Appendix-A matrix (`mack_shooting.mack_first_order_mat`) evaluated at $\beta = 0$ with $\lambda/\mu = 0$ (the same Stokes closure as Part A’s temporal solver), which at those settings is algebraically identical to the printed Eqs. 22–28 of [3], term for term, with one correction: the X_1 coefficient of the printed Eq. (24) reads $+i(2\mu \frac{d\mu}{dT} T' + \frac{4}{3} \frac{T'}{T})$, but re-deriving the pressure row from Eqs. (13) and (23) of the same reference gives $-i(2\frac{1}{\mu} \frac{d\mu}{dT} T' + \frac{4}{3} \frac{T'}{T})$ (sign and μ -placement). With the corrected term the two solvers agree in practice: at the worked-example parameters of Section 7, $\sigma_{\min}(c)$ falls to 4×10^{-6} at the spectral eigenvalue $c = 0.930 + 0.0200i$ and rises by two orders of magnitude within $|\Delta c| \sim 2 \times 10^{-3}$ of it.

12 Boundary conditions: the wall and the decaying freestream basis

Six conditions pin down a non-trivial disturbance: three homogeneous conditions at the wall, and three at the freestream selecting only the decaying solutions.

Wall conditions. At $y = 0$ the three primary amplitudes vanish,

$$X_1(0) = 0 \text{ (no slip)}, \quad X_3(0) = 0 \text{ (no penetration)}, \quad X_5(0) = 0 \text{ (isothermal wall)}, \quad (21)$$

the pointwise counterparts of the spectral solver’s wall identity rows (Section 3).

Freestream: the decaying basis. As $y \rightarrow y_{\max}$ the mean flow becomes uniform and the coefficient matrix approaches a constant, $A_{\infty} = A(y_{\max})$. A constant-coefficient system $DY = A_{\infty}Y$ has solutions $Y \propto e^{\lambda y}v$ built from the eigenpairs (λ, v) of A_{∞} ; the six eigenvalues split into three with $\text{Re } \lambda > 0$, which grow outward and are unphysical, and three with $\text{Re } \lambda < 0$, which decay outward as a bounded mode requires. Only the three decaying eigenvectors are retained (function `ozgen_freestream_decay_basis_2d`),

$$A_{\infty}v_k = \lambda_k v_k, \quad \text{Re}(\lambda_k) < 0, \quad k = 1, 2, 3, \quad (22)$$

assembled into a 6×3 matrix and orthonormalised by a QR factorisation into the starting basis $Y(y_{\max})$. Enforcing decay in this way is the shooting analogue of the spectral solver's freestream identity rows. If a trial c yields fewer than three strictly decaying eigenvalues, the code takes the three most decaying (smallest $\text{Re } \lambda$) so the march remains well defined; near the physical eigenvalue exactly three decay.

13 Integrating the bounded basis to the wall

The three admissible freestream solutions must be propagated to the wall so their arrival values can be tested against (21). Because three independent solutions are tracked simultaneously, the state is the 6×3 matrix Y whose columns are the three solutions. It is marched from $y_{\max}$ down to $y = 0$ on a fixed grid of about $n_{\text{steps}} = 800$ steps with the classical fourth-order Runge–Kutta scheme (function `integrate_ozgen_bounded_basis_2d`); over one step of size h (here $h < 0$),

$$\begin{aligned} k_1 &= A(y_0)Y, & k_2 &= A(y_m)(Y + \tfrac{1}{2}h k_1), & Y &\leftarrow Y + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4), \\ k_3 &= A(y_m)(Y + \tfrac{1}{2}h k_2), & k_4 &= A(y_1)(Y + h k_3), \end{aligned} \quad (23)$$

with $y_m = \frac{1}{2}(y_0 + y_1)$ the step midpoint and $A(\cdot)$ the pointwise matrix of Section 11.

Re-orthonormalisation. The three solutions decay at very different rates, so after a short march the rapidly decaying columns are numerically swamped by the slowest one and the basis collapses toward a single direction, the parasitic-error pathology classical in shooting for stiff stability problems [12, 1]. To suppress it, after every RK4 step the columns are re-orthonormalised by a QR factorisation,

$$Y \longleftarrow Q, \quad \text{where } Y = QR \quad (Q \in \mathbb{C}^{6 \times 3} \text{ orthonormal}), \quad (24)$$

retaining only the orthonormal factor; the discarded triangular factor merely records scaling. The operation preserves the *subspace* spanned by the columns while keeping them well conditioned, and it is that subspace, not the individual column magnitudes, on which the eigenvalue condition below depends.

14 The eigenvalue condition and the wall matrix

At the wall the integrated basis is a 6×3 matrix $Y(0)$, and any admissible disturbance is a combination $Y(0)a$ of its columns, $a \in \mathbb{C}^3$. The wall conditions (21) require rows 1, 3, 5 of that combination to vanish; collecting those rows gives the 3×3 *wall matrix* (function `ozgen_temporal_wall_matrix_2d`)

$$M(c) = \begin{bmatrix} Y_1(0) \\ Y_3(0) \\ Y_5(0) \end{bmatrix} \in \mathbb{C}^{3 \times 3}, \quad \text{wall conditions: } M(c)a = 0. \quad (25)$$

A non-trivial $a \neq 0$ exists only when $M(c)$ is singular. Rather than testing a determinant, which is poorly scaled after the repeated re-orthonormalisation, the code monitors the smallest singular value of $M(c)$ (function `ozgen_temporal_sigma_min_2d`),

$$\sigma_{\min}(c) = \min \text{svd}(M(c)), \quad \boxed{\sigma_{\min}(c) \rightarrow 0 \iff c \text{ is an eigenvalue}}, \quad (26)$$

a non-negative, well-scaled measure of distance to singularity: it vanishes exactly when the three decaying solutions can be combined to satisfy all three wall conditions, that is, exactly when a genuine boundary-layer mode exists at that c .

15 The eigenvalue search

It remains to locate the complex $c = c_r + ic_i$ at which $\sigma_{\min}(c) = 0$, starting from a physics-based guess; this single c is the temporal eigenvalue, identical to the one Part A returns. Since c is complex, driving $\sigma_{\min} \rightarrow 0$ is a two-dimensional real minimisation over (c_r, c_i) . `pyMack` applies the Nelder–Mead simplex method [13] (`scipy.optimize.minimize`, function `solve_ozgen_temporal_mode_2d_sigma_min`) to the objective $\sigma_{\min}$, from an initial guess such as the second-mode estimate $c_r \approx 1 - 1/M_e$. The search is confined to the physical strip

$$c_r \in (0, 1.2), \quad c_i \in (-0.2, 0.2), \quad (27)$$

enforced by a large penalty (10^6) outside the box, and converges to the tolerances

$$\text{xatol} = 10^{-7} \text{ (in } c), \quad \text{fato1} = 10^{-8} \text{ (in } \sigma_{\min}), \quad (28)$$

within at most ~ 120 iterations. Each objective evaluation runs the entire pipeline of Sections 12–14 (Figure 10): build the freestream decay basis, integrate with re-orthonormalisation to the wall, form $M(c)$, take $\sigma_{\min}$. The converged minimiser is the eigenvalue, and the growth rate follows exactly as in Part A, $\omega_i = \alpha c_i$. The five stages map to functions as follows:

step	operation	function
1	build the 6×6 pointwise matrix $A(y)$ from the mean flow	<code>ozgen_first_order_matrix_2d</code>
2	eigen-decompose $A(y_{\max})$; keep the 3 decaying modes; QR to a 6×3 basis	<code>ozgen_freestream_decay_basis_2d</code>
3	RK4-march the 6×3 basis $y_{\max} \rightarrow 0$, QR after every step	<code>integrate_ozgen_bounded_basis_2d</code>
4	collect wall rows (X_1, X_3, X_5) into $M(c)$; take $\sigma_{\min}(c)$	<code>ozgen_temporal_wall_matrix_2d</code> , <code>ozgen_temporal_sigma_min_2d</code>
5	Nelder–Mead over (c_r, c_i) until $\sigma_{\min} \rightarrow 0$; return c	<code>solve_ozgen_temporal_mode_2d_sigma_min</code>

16 Spectral versus shooting

The two solvers answer the identical question (for a given real α , at what complex c does a non-trivial disturbance decay at the freestream and satisfy the wall conditions?) and return the same physical phase speed, but with complementary characteristics, mirroring the classical boundary-value/initial-value trade-off [2]:

	Spectral (Part A)	Shooting (Part B)
core object	one $4n \times 4n$ pair (A, B)	one 6×6 matrix $A(y)$, evaluated pointwise
what it returns	<i>all</i> $4n$ eigenvalues at once	<i>one</i> eigenvalue near a guess
freestream decay	identity rows on the edge nodes	the 3 decaying eigenvectors of $A(y_{\max})$
wall conditions	identity row replacement	$M(c)$ singular, $\sigma_{\min}(c) \rightarrow 0$
how the answer is found	dense QZ / shift-invert	RK4 march + QR, then Nelder–Mead on $\sigma_{\min}$
initial guess	not required	required (physics-based)
memory / cost	large dense matrices, $O((4n)^3)$	tiny system, cheap per evaluation
best used for	surveying the spectrum; sweeping neutral curves and N -factors	pinpointing one mode to tight tolerance; independent verification

In practice the two are used together: the spectral solver locates a mode without prior knowledge of the spectrum, and the shooting solver refines it and confirms it through entirely independent numerics. Agreement between the two on c is strong evidence that the eigenvalue is physics rather than a discretisation artifact.

Code map (Part B). decay basis, integration, wall matrix, singular value, and search: `pymack/temporal.shooting.py`; the 6×6 matrix is shared with the Mack Appendix-A module (`pymack/mack.shooting.py`, `mack.first_order_matrix_6` at $\beta = 0$, $\lambda/\mu = 0$).

References

- [1] L. M. Mack, “Boundary-layer linear stability theory,” AGARD Report No. 709, *Special Course on Stability and Transition of Laminar Flow*, 1984.
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